WHAT IF WE COULD ASK A REPRESENTATIVE POPULATION QUESTIONS THAT WERE NEVER INCLUDED IN A SURVEY?

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SIMULATING COGNITIVE SECURITY WITH GENERATIVE (SYNTHETIC) AGENTS: TESSARON

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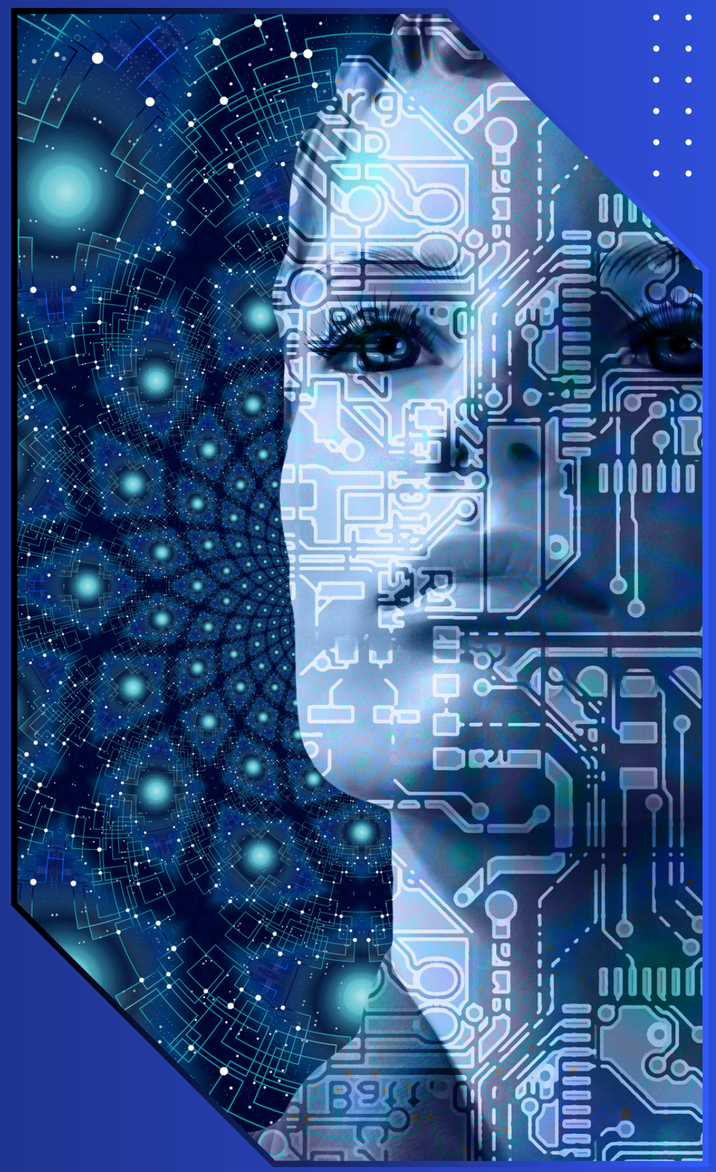

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TESSARON AGENTIC NETWORK: THE LONG-TERM GOAL

Tessera, a Latin word referring to a small tile or piece used in a mosaic.

Synthetic + cybersecurity - easing the way into much more sophisticated issues.

The research problem: can we study a population without asking its members?

Park et al. (2024), Rupprecht et al. (2026) - generative agents

Validation as the central challenge in the field (Larooij & Törnberg, 2026).

FROM GENERATIVE AGENTS TO SYNTHETIC POPULATIONS

Can language models reproduce human response patterns?

- Generative agents combine language models with structured information about individuals, context, and potentially memory.
- Silicon sampling conditions language-model responses on respondent characteristics.
- Algorithmic fidelity concerns correspondence between simulated and observed patterns of human responses.
- Existing research demonstrates both promising approximations and systematic distortions in synthetic survey data.

Argyle et al. (2023); Park et al. (2023); Bisbee et al. (2024)

INTRODUCING TESSARON

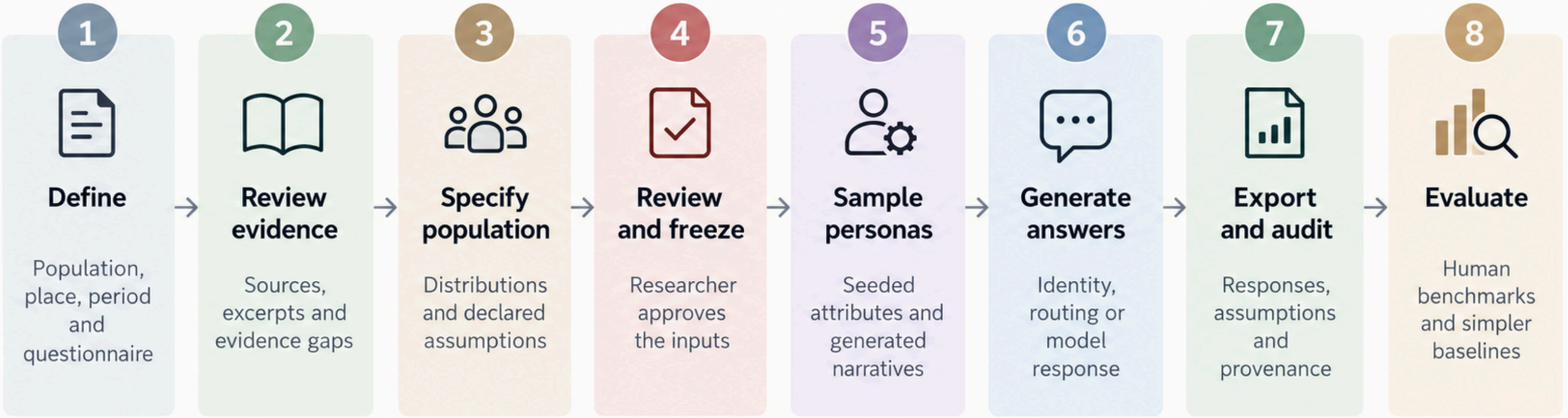
A modular framework for constructing and evaluating synthetic populations

- Constructs synthetic personas from quantitative population constraints and theoretically relevant characteristics.
- Integrates literature retrieval, population modelling, persona generation, and survey administration.
- Generates structured responses using language models.
- Evaluates synthetic response distributions against human survey benchmarks.
- Preserves researcher oversight and traceability across the workflow.

RQ: *can quantitatively constrained generative agents reproduce observed population-level response distributions?*

The Tessaron workflow

From evidence and assumptions to evaluation.



An inspectable workflow makes errors traceable. It does not establish empirical validity.

TESTING TESSARON AGAINST EUROBAROMETER 499

- Human benchmark: Czech Internet users aged 15+, October 2019.
- Synthetic cohort: 100 constructed personas.
- Ten cybersecurity-concern items (QC8_1–QC8_10).
- One multiple-response password-change question (QC4).
- Comparison with published, weighted Eurobarometer response percentages.
- Fixed synthetic cohort across response-generation configurations.

Evaluation: mean absolute category error, measured in percentage points.

Why use Tessaron if Eurobarometer already exists?

Eurobarometer is the benchmark. Tessaron is the tool for questions it cannot answer.

Use original Eurobarometer data when you need...

- Observed human responses
- Established survey relationships
- Representative estimates for asked questions
- The strongest evidence about what people actually reported

Use Tessaron when you need to...

- Explore questions never asked in the survey
- Test candidate scenarios, warnings, or interventions
- Reuse a fixed synthetic population across conditions
- Prioritize which ideas merit human data collection

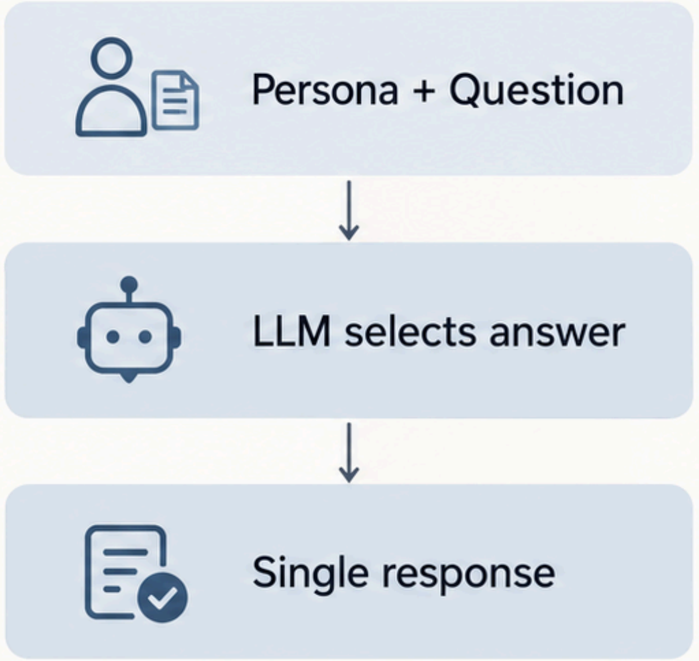
Key point: Tessaron does not replace Eurobarometer; it extends research beyond the original questionnaire.

Best used as a complement, not a substitute.

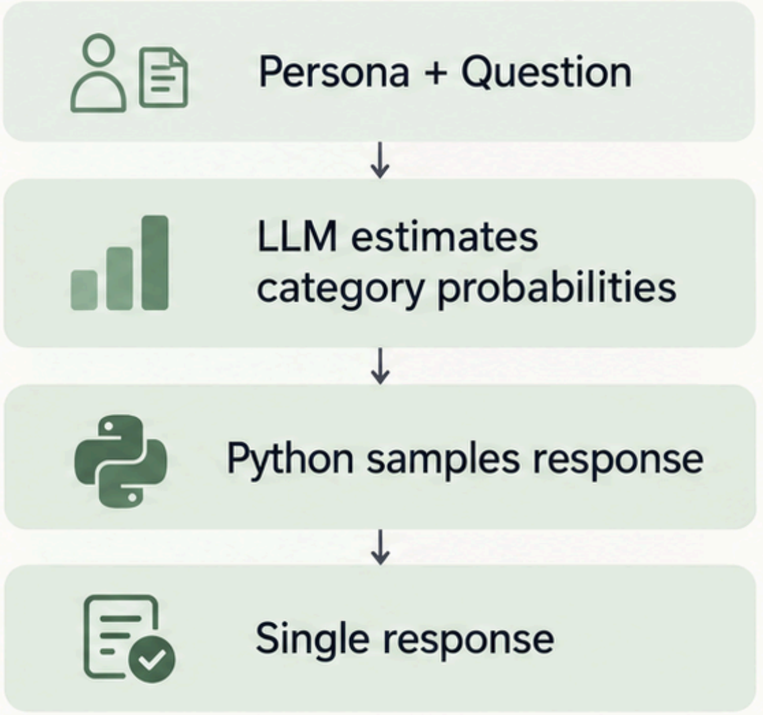
Does the Answering Procedure Matter?

Same synthetic personas. Different response-generation procedures.

Original procedure



Probability-based procedure



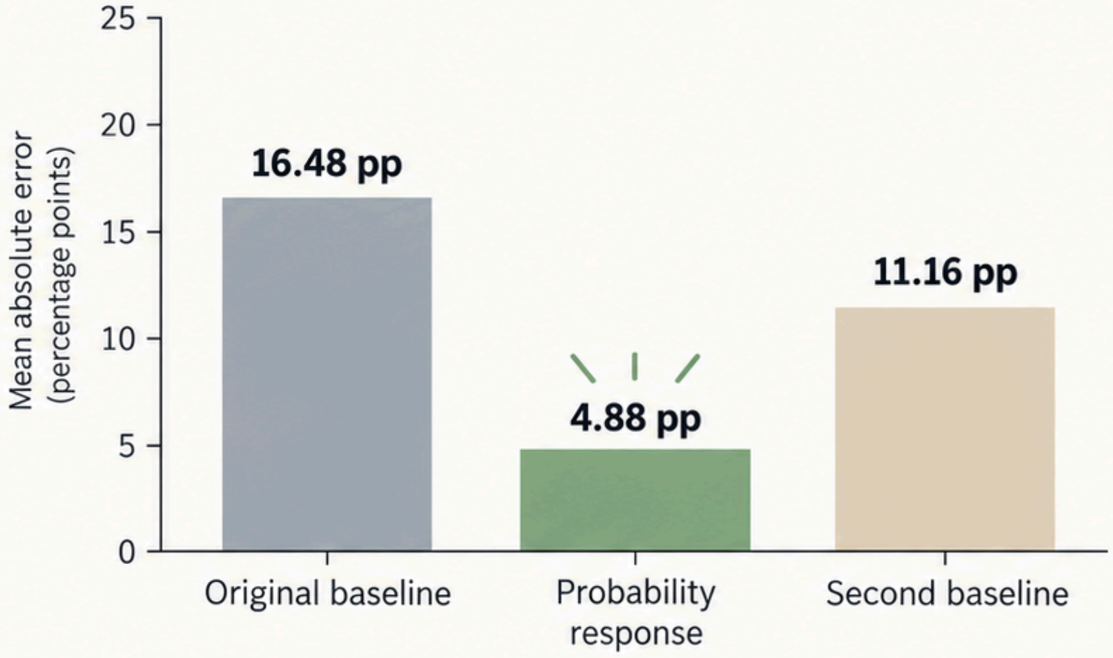
Core difference: Separating probability elicitation from categorical answer selection.

Tessaron: Error comparison

Mean absolute error in percentage points. Lower is better.

Cybercrime concern (QC8)

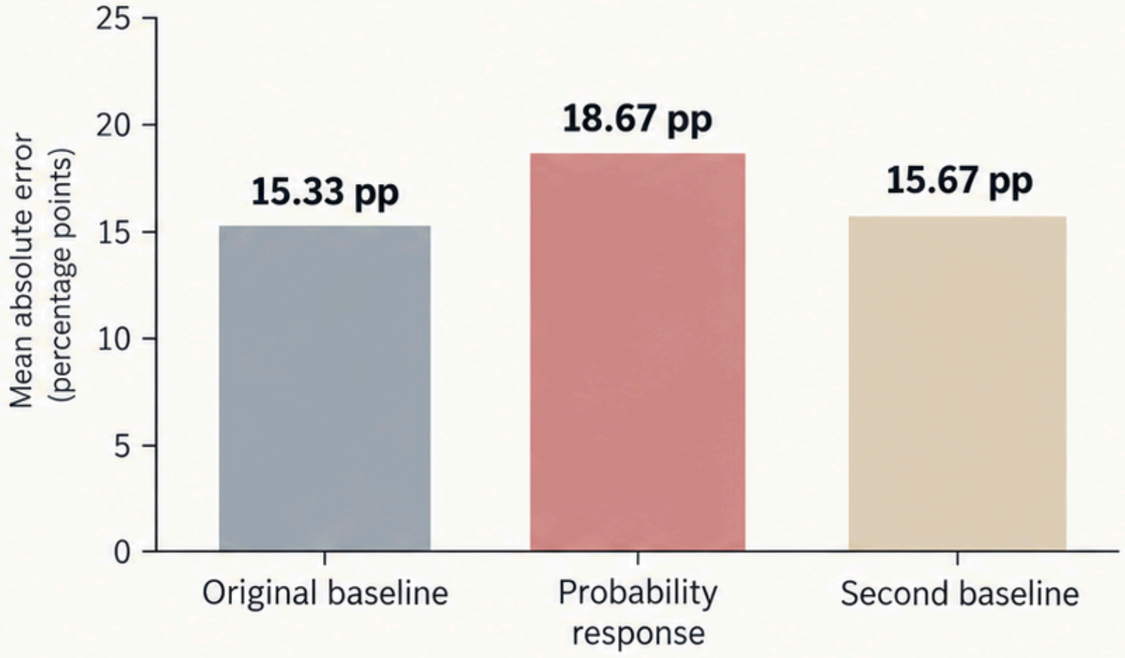
Mean of 10 item MAEs



~70% lower error
vs. original baseline

Password changes (QC4)

Mean category error



Attitudinal fit did not generalize
to self-reported behaviour

BUT FIDELITY DOES NOT GENERALIZE AUTOMATICALLY

Improved attitudinal correspondence ≠ accurate behavioural self-reports

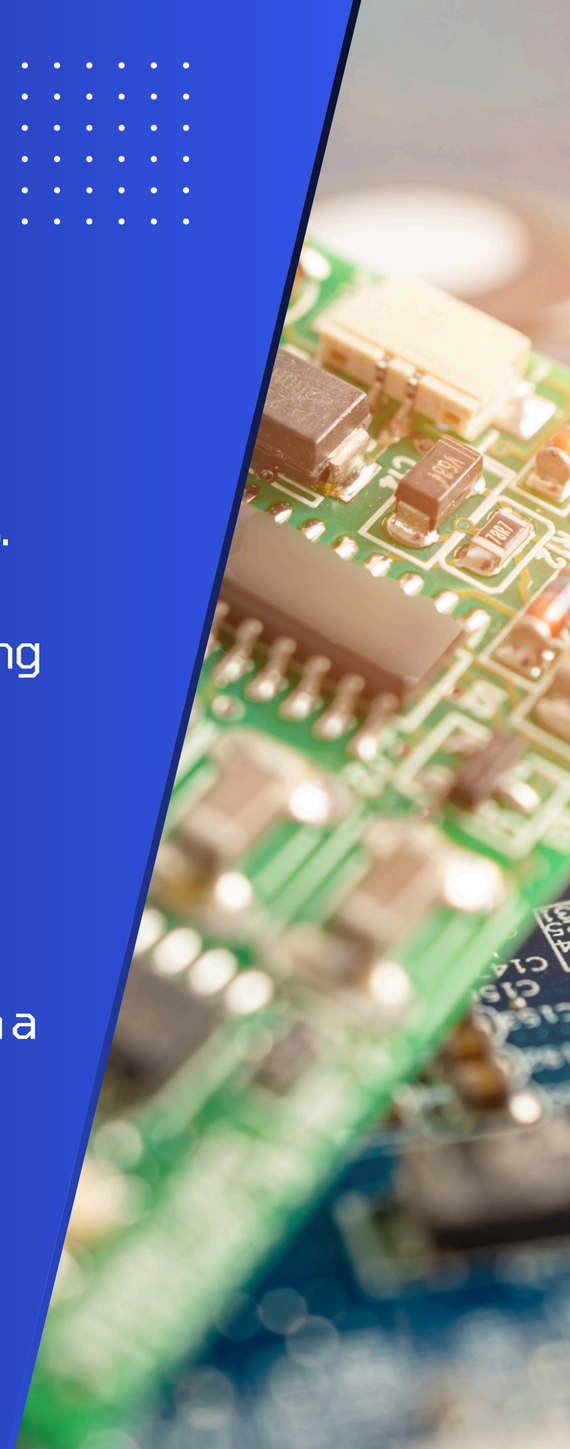
- Large discrepancies remain in reported password-changing practices.
- Accuracy varies across survey constructs and question formats.
- Matching aggregate attitudes does not establish behavioural, individual-level, or mechanistic fidelity.
- The password-change item retained direct generation in both configurations.

Implication: synthetic-population validity is outcome-specific and cannot be inferred from a single successful benchmark.

THE ROAD AHEAD — FROM FIT TO PREDICTION

Can Tessaron generalize to questions and people it has not seen?

- Hold out the test: evaluate unseen survey items, subgroup patterns and relationships.
- Measure robustness: repeat responses across fixed personas and models, quantifying variability.
- Test what adds value: compare different persona and response-generation configurations against simpler statistical models.
- Predict before observing: test predicting via experimental vignettes against data from a representative Czech survey - collected after locked-in predictions.



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THANK YOU FOR ATTENTION

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